

JR. PHYSICS QUESTIONS FOR IPE

S.No	TOPICS	8 M	4 M	2 M	TOTAL
1	PHYSICAL WORLD			2M	2
2	UNITS AND MEASUREMENTS			2M	2
3	MOTION IN A STRAIGHT LINE		4M		4
4	MOTION IN A PLANE		4M	2M	6
5	LAWS OF MOTION		4M	2M	6
6	WORK, ENERGY AND POWER	8M			8
7	SYSTEM OF PARTICLES & ROTATIONAL MOTION		4M +4M		8
8	OSCILLATIONS	8M			8
9	GRAVITATION		4M		4
10	MECHANICAL PROPERTIES OF SOLIDS		4M		4
11	MECHANICAL PROPERTIES OF FLUIDS			2M+2M	4
12	THERMAL PROPERTIES OF MATTER		4M	2M+2M	8
13	THERMODYNAMICS	8M			8
14	KINETIC THEORY			2M+2M	4
		24	32	20	76

LAQ'S

***Work Power Energy :**

1. Develop the notions of work and kinetic energy and show that it leads to Work-Energy Theorem?
2. State the law of conservation of energy and verify it in case of a freely falling body. What are the conditions under which the law of conservation of energy is applicable?
3. What are collisions? Explain the possible types of collisions? Develop the theory of one dimensional elastic collision?

***Oscillations :**

4. Define simple harmonic motion. Show that the motion of (point) projection of particle performing uniform circular motion, on any diameter, is simple harmonic.
5. Show that the motion of a simple pendulum is simple harmonic and hence derive an equation for its time period. What is second's pendulum?

- * Derive equation for the kinetic energy and potential energy of a simple harmonic oscillator and show that the total energy of a particle in simple harmonic motion is constant at any point on its path.

***Thermodynamics:**

6. Explain reversible and irreversible processes. Describe the working of Carnot engine. Obtain an expression for the efficiency.
7. State second law of thermodynamics. How is heat engine different from a refrigerator?

SAQ's

1. Motion in a Straight line:

1. A particle moves in a straight line with uniform acceleration. Its velocity at time $t = 0$ is V_1 and at time $t = t$ is V_2 . Average velocity of the particle in this time interval is $\frac{V_1 + V_2}{2}$. Is this statement true or false? Substantiate your answer?
2. Derive $s = ut + \frac{1}{2}at^2$ using Graph method? **(or)**
Derive the equation $x = V_0t + \frac{1}{2}at^2$ using graphical method?
3. Explain the terms 'average velocity' and 'instantaneous velocity'. When are they equal?
- * Can the velocity of an object be in a direction other than the direction of acceleration of the object? Explain
- * A ball is dropped from the roof of a tall building and simultaneously another ball is thrown horizontally with some velocity from the same roof. Which ball lands first? Explain your answer.

2. Motion in a Plane:

4. State parallelogram law of vectors. Derive an expression for the magnitude and direction of the resultant vector.
5. Show that the trajectory of an object thrown at certain angle with the horizontal is a parabola.

6. Show that the maximum height and range of a projectile are $\frac{u^2 \sin^2 \theta}{2g}$ and $\frac{u^2 \sin 2\theta}{g}$ respectively, where the terms have their regular meaning.
7. Define unit vector, null vector and position vector.
8. If $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$ prove that the angle between $\vec{a}$ and $\vec{b}$ is 90°
9. Show that for a projectile launched at an angle 45° , the maximum height of the projectile is one quarter of the range ?

3.Laws of Motion:

10. Define the terms momentum and impulse. State and explain the law of conservation of linear momentum. Give its examples.
11. Explain the advantages and the disadvantages of friction.
12. Mention the methods used to decrease friction.
13. State Newton's second law of motion. Hence derive the equation of motion $F=ma$ from it.
14. Why is pulling the lawn roller is preferred than pushing the lawn roller?
- * Explain the terms limiting friction, dynamic friction and rolling friction.
- * Why are shock absorbers used in motor cycles and cars? Explain.
- * State the laws of rolling friction.

4. System of Particles & Rotational Motion :

15. Distinguish between centre of mass and centre of gravity.
16. Define vector product. Explain the properties of a vector product with two examples.
17. Define angular velocity (ω). Derive $v = r\omega$
18. Define angular acceleration and torque. Establish the relation between angular acceleration and torque.
- * Explain about the centre of mass of Earth-Moon system and its rotation around the sun.
- * Write the equations of motion for a particle rotating about a fixed axis?

5. Gravitation :

19. State Kepler's laws of planetary motion?
20. What is orbital velocity? Obtain an Expression for it.
21. What is escape velocity? Obtain an expression for it.
22. What is Geostationary satellite. Write it uses.
- * Derive the relation between acceleration due to gravity (g) at the surface of a planet and gravitational constant (G).

6. Mechanical Properties of Solids:

23. Describe the behavior of a wire under gradually increasing load?
24. Define modulus of elasticity, stress, strain and Poisson's ratio.
25. Define stress and explain the types of stress.
26. Define strain and explain the types of strain.
27. Define strain energy and derive the equation for the same.
- * Define Hooke's law of elasticity, proportionality limit, permanent set, and breaking stress.
- * Define Young's modulus, Bulk modulus and rigidity modulus.

7. Thermal Properties of Matter :

28. Pendulum clocks generally go fast in winter and slow in summer. Why?
29. In what way is the anomalous behavior of water advantageous to aquatic animals?
30. Explain conduction, convection and radiation with examples.
31. Write a short note on triple point of water.
32. State Boyle's law and Charles's law. Derive ideal gas equation.
- * Explain Celsius and Fahrenheit scales of temperature. Obtain the relation between Celsius and Fahrenheit scales of temperature.
- * Two identical rectangular strips one of copper and the other of steel are riveted together to form a compound bar (bimetallic strip). What will happen on heating?

Work Power Energy

- * What is potential energy? Derive an expression for it.
- * A lorry and a car moving with the same momentum are brought to rest by the application of brakes, which provide equal retarding forces. Which of them will come to rest in shorter time? Which will come to rest in less distance?
- * Distinguish between conservative and non-conservative forces with one example each.

VSAQ's

1.Physical World:

1. What is physics?
2. What is the discovery of C.V.Raman.
3. What are the fundamental forces in nature?
4. What is contribution of S. Chandra Sekhar to Physics?
- * Which of the following has symmetry?
 - a) Acceleration due to gravity.
 - b) Law of gravitation

2. Units and Measurements:

5. Distinguish between accuracy and precision.
6. How can systematic errors be minimized or eliminated?
7. Distinguish between fundamental units and derived units.
8. Why do we have different units for the same physical quantity?
- * Express unified atomic mass unit in kg.

3.Motion in a Plane:

9. The vertical component of a vector is equal to its horizontal component. What is the angle made by the vector with x-axis?
10. Two forces of magnitudes 3 units and 5 units act at 60° with each other. What is the magnitude of their resultant?
11. $\vec{A} = \vec{i} + \vec{j}$, what is the angle between vector and x-axis.
12. When two right angled vectors of magnitude 7 units and 24 units combine what is the magnitude of their resultant?
13. If $\vec{P} = 2\vec{i} + 4\vec{j} + 14\vec{k}$ and $\vec{Q} = 4\vec{i} + 4\vec{j} + 10\vec{k}$, find the magnitude of $\vec{P} + \vec{Q}$.
14. What is the acceleration of a projectile at the top of its trajectory?

- * A vector $\vec{v}$ makes an angle θ with the horizontal. The vector is rotated through an angle α . Does this rotation change the vector $\vec{v}$?
- * Can a vector of magnitude zero have nonzero components?

4. Laws of Motion:

15. A horse has to pull harder during the start of the motion than later. Explain?
16. What is inertia? What gives the measure of inertia?
17. According to Newton's third law, every force is accompanied by an equal and opposite force. How can a movement ever take place?
18. When a bullet is fired from a gun, the gun gives a kick in the backward direction. Explain.
19. If a bomb at rest explodes into two pieces, the pieces must travel in opposite directions. Explain.
20. Can the coefficient of friction be greater than one?
21. Why does the car with a flattened tyre stop sooner than the one with inflated tyres?
22. What happens to coefficient of friction if weight of the body is doubled?
- * Define force. What are the basic forces in nature?

5. Mechanical Properties of fluids.:

23. Define average pressure. Mention its unit and dimensional formula. Is it a scalar or a vector?
24. Define Viscosity. What are its units and dimension?
25. What is Magnus effect?
26. What is the principle behind the carburetor of an automobile?
27. Why are drops and bubbles spherical?
28. Give the expression for the excess pressure in a liquid drop.
29. Give the expression for the excess pressure in an air bubble inside the liquid.
30. Give the expression for the excess pressure in a soap bubble in air.
31. What are water proofing agents and water wetting agents? What do they do?
32. What is angle of contact? What are the values for pure water and mercury.

6. Thermal Properties of Matter:

33. Distinguish between heat and temperature.
34. What are the lower & upper fixed points in Celsius & Fahrenheit scales?
35. Can a substance contract on heating? Give examples.
36. Why gaps should be left between two successive rails?
37. What is latent heat of vapourisation?
38. What are the units and dimensions of specific gas constant?
39. State Wein's displacement law.
40. Ventilators are provided in rooms just below the roof. Why?
41. What is greenhouse effect? Explain global warming?
42. Define absorptive power of a body. What is the absorptive power of a perfect black body?
43. State Newton's law of cooling.
44. The roof buildings are often painted, white during summer. Why?
- * Why do liquids have no linear and areal expansions?
- * What is latent heat of fusion?
- * What is specific gas constant? Is it same for all gases?
- * Why utensils are coated black? Why the bottom of the utensils are made of copper?
- * Define temperature gradient and coefficient of thermal conductivity.
- * What is Regelation of ice?

7. Kinetic Theory of Gases:

45. Define mean free paths.
46. When does a real gas behave like an ideal gas?
47. State Boyle's Law and Charles Law
48. State Dalton's law of partial pressures.
49. Pressure of an ideal gas in container is independent of shape of the container – explain
50. What is the expression between pressure and kinetic energy of a gas molecule?
- * Name two prominent phenomena which provide conclusive evidence of molecular motion.
- * Explain the concept of degrees of freedom for molecules of a gas.
- * If the absolute temperature of a gas is increased to 3 times, what will be the increase in rms velocity of the gas molecule?
- * What is the law of equipartition of energy?

Work power energy:

- * State the conditions under which a force does no work.
- * Which physical quantity remains constant
(a) in elastic collision (b) in inelastic collision
- * A body freely falling from a certain height 'h' after striking a smooth floor rebounds and rises to a height h/2. What is the coefficient of restitution between the floor and the body?

Sytem of Particles & Rotational Motion:

- * Is it necessary that a mass should be present at the centre of mass of any system?
- * Why are spokes provided in a bicycle wheel?
- * By spinning eggs on a table top, how will you distinguish a hard boiled egg from a raw egg?
- * We cannot open or close the door by applying force at the hinges. Why?
- * Why should a helicopter necessarily have two propellers?
- * If the polar ice caps of the earth were to melt, what would the effect of the length of the day be?
- * Why do we prefer a spanner of longer arm as compared to the spanner of shorter arm?

Problems :

1. Motion in a Straight Line :

1. A car travels the first third of a distance with a speed of 10kmph, the second third at 20 kmph and the last third of 60 kmph. What is its mean speed over the entire distance?
2. A man walks on a straight road from his home to a market 2.5 km away with a speed of 5 km h^{-1} . Finding the market closed, he instantly turns and walks back home with a speed of 7.5 km h^{-1} . What is the (a) magnitude of average velocity and (b) average speed of the man over the time interval 0 to 50 min.

2. Motion in a Plane:

3. A force $2\vec{i} + \vec{j} - \vec{k}$ Newton's acts on a body which is initially at rest. At the end of 20 seconds, the velocity of the body is $4\vec{i} + 2\vec{j} - 2\vec{k}$ m/s. What is the mass of the body?

4. A projectile is fired at an angle of 60° to the horizontal with an initial velocity of 800 m/s
 - (i) Find the time of flight of the projectile before it hits the ground
 - (ii) Find the distance it travels before it hits the ground (range)
 - (iii) Find the time of flight for the projectile to reach its maximum height.
5. O is a point on the ground chosen as origin. A body first suffers a displacement of $10\sqrt{2}$ m North – East, next 10 m North and finally $10\sqrt{2}$ m North – West. How far it is from the origin?
6. Wind is blowing from the south at 5 ms^{-1} . To a cyclist it appears to be blowing from the east at 5 ms^{-1} . Find the velocity of the cyclist.

3.Laws of Motion :

7. Calculate the time needed for a net force of 5 N to change the velocity of a 10 kg mass by 2 m/s.
8. A constant force acting on a body of mass 3.0 kg changes its speed from 2.0 ms^{-1} to 3.5 m/s in 25 s. The direction of motion of the body remains unchanged. What is the magnitude and direction of the force?
9. A man in a lift feels an apparent weight W when the lift is moving up with a uniform acceleration of $\frac{1}{3}$ rd of the acceleration due to gravity. If the same man were in the same lift now moving down with a uniform acceleration that is $\frac{1}{2}$ of the acceleration due to gravity, then what is his apparent weight?

4.Work Power – energy :

10. A machine gun fires 360 bullets per minute and each bullet travels with a velocity of 600 ms^{-1} . If the mass of each bullet is 5 gm, find the power of the machine gun?
11. Find the useful power used in pumping 3425 m^3 of water per hour from a well 8m deep to the surface, supposing 40% of the horse power during pumping is wasted. What is the horse power of the engine?
12. A pumps required to lift 600 kg of water per minute from a well 25 m deep and to eject it with a speed of 50 ms^{-1} . Calculate the power required to perform the above task?

13. A ball falls from a height of 10 m on to a hard horizontal floor and repeatedly bounces. If the coefficient of restitution is $\frac{1}{\sqrt{2}}$, What is the total distance travelled by the ball before it ceases to rebound?
14. A body constrained to move along the z-axis of a coordinate system is subject to a constant force (F) given by $F = \hat{i} + 2\hat{j} + 3\hat{k}$ N where $\hat{i}, \hat{j}, \hat{k}$ are unit vectors along the x-, y- and z-axis of the system respectively. What is the work done by this force in moving the body a distance of 4m along the z-axis?
15. Find the total energy of a body of 5 kg mass, which is at a height of 10 m from the earth and falling downwards straightly with velocity of 20 m/s (Take $g = 10 \text{ m/s}^2$)
16. A batsman hits back a ball straight in the direction of the bowler without changing its initial speed of 12 ms^{-1} . If the mass of the ball is 0.15 kg, determining the impulse imparted to the ball (Assume linear motion of the ball)
17. Calculate the power of the pump required to lift 600 kg of water per minute from a well of 25 m deep.

5. Sytem of Particles & Rotational Motion:

18. Find the torque a force $7\hat{i} + 3\hat{j} - 5\hat{k}$ about the origin. The forces acts on a particle whose position vector is $\hat{i} - \hat{j} + \hat{k}$
19. Find the centre of mass of three particles at the vertices of an equilateral triangle. The masses of the particle masses of the particles are 100 g, 150 g and 200 g respectively, Each side of the equilateral triangle is 0.5 m long.
20. Obtain the centre of mass of a system of 4 particles 1 kg, 2 kg, 3 kg, and 4 kg respectively located at the corners of square of 1m side. Take origin at 1 kg, 2 kg, and 3 kg are on X and Y-axis.
21. Find the position of centre of mass of the system of 3 objects of masses 1 kg, 2 kg, 3 kg located at the corners of an equilateral triangle of side 1m. Take 1 kg mass object at the origin and 2 kg is along X-axis.
22. If the centre of mass of three particles of masses 1 kg, 2 kg, 3 kg, is at (2, 2, 2), then where should a fourth particle of mass 4 kg will be placed so that the combined centre of mass be at (0, 0, 0).

23. When 100 J of work is done on a fly wheel, its angular velocity is increased from 60 rpm to 180 rpm. What is the moment of inertia of the wheel?
24. Find the scalar and vector products of two vectors $a = 3\hat{i} - 4\hat{j} + 5\hat{k}$ and $b = -2\hat{i} + 4\hat{j} - 5\hat{k}$.
25. The moment of inertia of a fly wheel making 300 revolutions per minute is 0.3 kgm^2 . Find the torque required to bring it to rest in 20s.

6. Mechanical Properties of Fluids:

26. If the diameter of a soap bubble is 10 mm and its surface tension is 0.04 Nm^{-1} , find the excess pressure inside the bubble.
27. What is the pressure on a swimmer 10 m below the surface of a lake?

7. Thermal Properties of Matter

28. What is the temperature for which the readings on Kelvin's Fahrenheit scales are same?
29. Find the increase in temperature of aluminum rod if its length is to be increased by 1% (α for aluminum = $25 \times 10^{-6}/^\circ\text{C}$)
30. The mass of a liter of gas is 1.562 g at 0°C under a pressure of 76 cm of mercury. The temperature is increased to 250°C and the pressure to 78 cm of mercury. What is the mass of one liter of the gas under new conditions?
31. The volume of a mass of gas at 37°C and a pressure of 75 cm of mercury is 620 c.c. Find the volume at N.T.P
32. A body cools from 60°C to 40°C in minutes. What will be its temperature after next 7 minutes if the temperature of its surroundings is 10°C ?
33. A body cools from 80°C to 50°C in minutes. Calculate the time it takes to cools from 60°C to 30°C . The temperature of the surroundings is 20°C .
34. If the maximum intensity of ration for a black body is found at $1.45 \mu\text{m}$. What is the temperature of a radiating body (Wien's constat = $2.9 \times 10^{-3} \text{ mK}$)
35. A body cools down from 60°C to 50°C in 5 minutes and to 40°C in another 8 minutes. Find the temperature o the surroundings.

36. A copper bar of thermal conductivity 401 W/mK has one end at 104°C and the other end at 24°C . The length of the bar is 0.10m and the cross-sectional area is $1.0 \times 10^{-6} \text{ m}^2$. What is the rate of heat conduction, along the bar?

8. Thermodynamics :

37. A refrigerator is to maintain eatables kept inside at 90°C . If room temperature is 36°C , calculate the coefficient of performance.

9. Kinetic Theory of Gases :

38. The absolute temperature of a gas is increased 3 times. What will be the increase in r.m.s velocity of the gas molecule?
39. What is the ratio of r.m.s speed of Oxygen and Hydrogen molecules at the same temperature?
40. Four molecules of a gas have speeds 1,2,3 and 4km/s . Find rms speed of the gas molecules.